TECHNICAL PRESENTATION

DSA0202 – COMPUTER VISION WITH OPENCV

PRESENTED BY :
NAVADEEP R(192424110)

GUIDED BY:
Dr. MAYURANATHAN M

INTRODUCTION

- Face recognition is a computer vision technology that identifies or verifies a person using their facial features.
- Airport security systems use face recognition to quickly verify passengers and improve safety.
- OpenCV is an open-source computer vision library used for image processing and face detection.
- The system captures a passenger's image, processes it, and compares it with a stored database for identification.
- This technology provides faster, more accurate, and contactless passenger verification while reducing manual effort and improving security.

IMAGE PROCESSING

Definition

Image processing is the technique of improving and analyzing images so that a computer can detect and recognize faces accurately.

Steps in Image Processing

- **Image Capture:** The camera captures the passenger's face.
- **Preprocessing:** Converts the image to grayscale and removes noise.
- **Face Detection:** Identifies the location of the face in the image.
- **Feature Extraction:** Extracts unique facial features such as eyes, nose, and mouth.
- **Face Matching:** Compares the extracted features with the stored database to verify identity.

VISION LEVELS

Definition

Vision levels describe how a computer understands and interprets visual information.

Types of Vision Levels

- **Low-Level Vision:** Performs basic image processing like noise removal and edge detection.
- **Mid-Level Vision:** Detects and separates important objects, such as the passenger's face.
- **High-Level Vision:** Recognizes the person's identity and makes decisions, such as granting or denying access.

CAMERA MODELS

Definition

A camera model explains how a real-world scene is captured and converted into a digital image.

Role of Camera Models

- Captures clear facial images.
- Maintains accurate image perspective.
- Uses camera calibration for precise measurements.
- Handles different viewing angles and distances.
- Improves the accuracy of face recognition.

WORK FLOW

Working of the Face Recognition System

Process

- Passenger arrives at the airport.
- Camera captures the passenger's face.
- Image processing enhances the captured image.
- Face detection identifies the face.
- Facial features are extracted.
- Features are compared with the stored database.
- Identity is verified.
- Access is granted or a security alert is generated.

ADVANTAGES & CHALLENGES

Advantages

- Faster passenger verification.
- Improved airport security.
- Contactless identification.
- High accuracy and real-time processing.
- Reduces manual work and human errors.

Challenges

- Poor lighting conditions.
- Face masks or sunglasses may affect recognition.
- Different face angles can reduce accuracy.
- Privacy and data security concerns.
- Requires a well-maintained database.

CONCLUSION

Face recognition is an important application of computer vision in airport security.Image processing prepares the captured image for accurate analysis. Vision levels help the system detect, understand, and recognize faces. Camera models ensure high-quality image capture for reliable recognition. Together with OpenCV, these technologies provide a fast, secure, and efficient airport security system.

THANK YOU